

Conditional Probability Distribution

$P\{X = x_i / Y = y_j\} = \frac{P\{X = x_i, Y = y_j\}}{P\{Y = y_j\}} = \frac{p_{ij}}{p_{*j}}$ is called the conditional probability function of X , given that $Y = y_j$.

Another way of expressing (in terms of marginal distribution) is

$$f(x|y) = \frac{f(x, y)}{h(y)}, \text{ provided } h(y) > 0.$$

Similarly, the conditional distribution of Y given that $X = x$ is

$$f(y|x) = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0.$$

If we wish to find the probability that the discrete random variable X falls between a and b when it is known that the discrete variable $Y = y$, we evaluate

$$P(a < X < b \mid Y = y) = \sum_{a < x < b} f(x|y),$$

where the summation extends over all values of X between a and b . When X and Y are continuous, we evaluate

$$P(a < X < b \mid Y = y) = \int_a^b f(x|y) \, dx.$$

Exercise: Use the following table to find the conditional distribution of X , given that $Y = 1$, and use it to determine $P(X = 0 \mid Y = 1)$.

$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Answer: We need to find $f(x|y)$, where $y = 1$. First, we find that

$$h(1) = \sum_{x=0}^2 f(x, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}.$$

Now

$$f(x|1) = \frac{f(x, 1)}{h(1)} = \left(\frac{7}{3}\right) f(x, 1), \quad x = 0, 1, 2.$$

Therefore,

$$f(0|1) = \binom{7}{3} f(0, 1) = \binom{7}{3} \left(\frac{3}{14} \right) = \frac{1}{2}, \quad f(1|1) = \binom{7}{3} f(1, 1) = \binom{7}{3} \left(\frac{3}{14} \right) = \frac{1}{2},$$

$$f(2|1) = \binom{7}{3} f(2, 1) = \binom{7}{3} (0) = 0,$$

and the conditional distribution of X , given that $Y = 1$, is

x	0	1	2
$f(x 1)$	$\frac{1}{2}$	$\frac{1}{2}$	0

Finally,

$$P(X = 0 \mid Y = 1) = f(0|1) = \frac{1}{2}.$$

Exercise:

Given the joint density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, \ 0 < y < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

find $g(x)$, $h(y)$, $f(x|y)$, and evaluate $P(\frac{1}{4} < X < \frac{1}{2} | Y = \frac{1}{3})$.

Answer:

By definition of the marginal density. for $0 < x < 2$,

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) \, dy = \int_0^1 \frac{x(1+3y^2)}{4} \, dy \\ &= \left(\frac{xy}{4} + \frac{xy^3}{4} \right) \Big|_{y=0}^{y=1} = \frac{x}{2}, \end{aligned}$$

and for $0 < y < 1$,

$$\begin{aligned} h(y) &= \int_{-\infty}^{\infty} f(x, y) \, dx = \int_0^2 \frac{x(1 + 3y^2)}{4} dx \\ &= \left(\frac{x^2}{8} + \frac{3x^2 y^2}{8} \right) \bigg|_{x=0}^{x=2} = \frac{1 + 3y^2}{2}. \end{aligned}$$

Therefore, using the conditional density definition, for $0 < x < 2$,

$$f(x|y) = \frac{f(x, y)}{h(y)} = \frac{x(1 + 3y^2)/4}{(1 + 3y^2)/2} = \frac{x}{2},$$

and

$$P\left(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3}\right) = \int_{1/4}^{1/2} \frac{x}{2} dx = \frac{3}{64}.$$

Mathematical Expectation:

The *average* value of a random phenomenon is also termed as its *mathematical expectation* or *expected value*.

The expected value of a discrete r.v. is a weighted average of all possible values of the r.v., where the weights are the probabilities associated with the corresponding values.

For a discrete r.v. $\mathbf{X}$ with p.m.f. $\mathbf{p}(\mathbf{x})$,

$$E(X) = \sum_x x p(x)$$

For a continuous r.v. $\mathbf{X}$ with p.d.f. $\mathbf{f}(\mathbf{x})$,

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Expected Value of a function of a random variable

Consider a r.v. X with *p.d.f.* (*p.m.f.*) $f(x)$ and distribution function $F(x)$.

If $g(\cdot)$ is a function such that $g(X)$ is a r.v. and $E[g(X)]$ exists, then

$$E[g(X)] = \sum_x g(x) p(x) \quad (\text{for discrete r.v.})$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx \quad (\text{for continuous r.v.})$$

Particular cases:

1. If we take $g(X) = X^r$, r - being a positive integer, then

$$E[X^r] = \int_{-\infty}^{\infty} x^r f(x) dx$$

$$E[X^r] = \sum_x x^r p(x)$$

Which is defined as μ'_r , the r^{th} **moment (about origin)** of the probability distribution. Thus

$$\mu'_r(\text{ about origin}) = E (X^r)$$

In particular, if $r = 1$ μ'_1 (about origin) = $E (X) = \bar{X} = \mathbf{Mean}$

$$r = 2 \quad \mu'_2 (\text{ about origin}) = E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

2. If $g(X) = [X - E(X)]^r = [X - \bar{x}]^r$, then

$$E[X - E(X)]^r = \int_{-\infty}^{\infty} [x - E(X)]^r f(x) dx = \int_{-\infty}^{\infty} [x - \bar{x}]^r f(x) dx$$

$$E[X - E(X)]^r = \sum_x [x - E(X)]^r p(x) = \sum_x [x - \bar{x}]^r p(x)$$

which is μ_r , the r^{th} **moment about mean**.

if $r = 2$, we get $\mu_2 = E[X - E(X)]^2 = \int_{-\infty}^{\infty} [x - \bar{x}]^2 f(x) dx = \mathbf{Variance}$

3. If $g(x) = \text{constant} = c$,

$$E(c) = \int_{-\infty}^{\infty} c f(x) dx = c \int_{-\infty}^{\infty} f(x) dx = c$$

$$E(c) = \sum_x c p(x) = c \sum_x p(x) = c$$

Expected Value of Two-dimensional RV

If (X, Y) is a two-dimensional RV with joint *p.d.f.* $f(x, y)$ or joint *p.m.f.* p_{ij} , then

$$E[g(X, Y)] = \sum_i \sum_j g(x_i, y_j) p_{ij} \quad (\text{for discrete r.v.})$$

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) \, dx dy \quad (\text{for continuous r.v.})$$

Properties of Expectation:

1. **Addition theorem of Expectation:** $E(X + Y) = E(X) + E(Y)$

Proof:

$$\begin{aligned} E(X + Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f(x, y) \, dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x) f(x, y) \, dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y) f(x, y) \, dx dy \end{aligned}$$

$$E(X + Y) = E(X) + E(Y)$$

Generalisation: $E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$

or $E(\sum_{i=1}^n X_i) = \sum_{i=1}^n E(X_i)$

2. Multiplication theorem of Expectation:

If X and Y are independent random variables, then $E(XY) = E(X) \cdot E(Y)$

$$\begin{aligned} E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (xy) f(x, y) \, dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (xy) f(x) f(y) dx dy \quad [\text{Since, } X \text{ and } Y \text{ are independent}] \\ &= \left[\int_{-\infty}^{\infty} x f(x) \, dx \right] \left[\int_{-\infty}^{\infty} y f(y) \, dy \right] \end{aligned}$$

$$E(XY) = E(X) E(Y)$$

Generalisation: If $X_1, X_2, \dots, X_n$ are n - independent r.v.'s, then

$$E(X_1 \cdot X_2 \cdot \dots \cdot X_n) = E(X_1) \cdot E(X_2) \cdot \dots \cdot E(X_n)$$

or
$$E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n E(X_i)$$

3. If $\mathbf{X}$ is a random variable and $\mathbf{a}$ and $\mathbf{b}$ are constants, then

$$\mathbf{E}(\mathbf{aX} + \mathbf{b}) = \mathbf{a} \mathbf{E}(\mathbf{X}) + \mathbf{b}$$

Proof: $E(ax + b) = \int_{-\infty}^{\infty} (ax + b) f(x) dx$

$$= a \int_{-\infty}^{\infty} x f(x) dx + b \int_{-\infty}^{\infty} f(x) dx$$

$$E(ax + b) = a E(X) + b$$

If $b = 0$, then we get $E(aX) = a E(X)$

However,

$$E\left(\frac{1}{X}\right) \neq \frac{1}{E(X)}$$

$$E(X^{1/2}) \neq [E(X)]^{1/2}$$

$$E(\log X) \neq \log E(X)$$

$$E(X^2) \neq [E(X)]^2$$

Variance:

$$\text{Variance} = \text{Var}(X) = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$$

1. If X is a random variable, then $V(aX + b) = a^2 V(X)$

Proof:

Let $Y = aX + b$

$$E(Y) = E(aX + b)$$

$$= a E(X) + b$$

$$Y - E(Y) = (aX + b) - (a E(X) + b)$$

$$= a[X - E(X)]$$

$$E[Y - E(Y)]^2 = a^2 E[X - E(X)]^2$$

$$V(Y) = a^2 V(X)$$

2. $V(b) = 0$, *Variance of a constant is zero.*

3. $\sigma = \sqrt{\text{Variance}} = \sqrt{\text{Var}(X)}$ = Standard Deviation.

Example 1

Let X be a random variable with the following probability distribution:

X	:	-3	6	9
$P(X = x)$	:	1/6	1/2	1/3

Find the values of $E(X)$, $E(X^2)$ and $E(2X + 1)^2$

$$* E(X) = \sum_x x p(x)$$

$$* E(X^2) = \sum_x x^2 p(x)$$

$$* E(2X+1)^2 = E(4X^2 + 4X + 1) = 4E(X^2) + 4E(X) + 1$$

$$\text{Ans. } E(X) = 11/2, E(X^2) = 93/2, E(2X+1)^2 = 209.$$

Example 2

Find the expectation of the number on a die when thrown.

Answer:

X	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E(X) = \sum x_i p_i$$

$$= 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6}$$

$$= \frac{1+2+3+4+5+6}{6}$$

$$= \frac{21}{6} = \frac{7}{2}$$

Example 3:

Two unbiased dice are thrown. Find the expected values of the sum of numbers of points on them.

Answer:

Let "x" indicate the sum of the points on the two dice
[Since you are required to find the expected value the sum of the scores on the two dice, the variable would represent the sum of the points on the dice]

The sum of the points on the two dice would be

- 2 if the dice show up {(1,1)}
- 3 if the dice show up {(1,2),(2,1)}
- 4 if the dice show up {(1,3),(2,2),(3,1)}
- 5 if the dice show up {(1,4),(2,3),(3,2),(4,1)}
- 6 if the dice show up {(1,5),(2,4),(3,3),(4,2),(5,1)}
- 7 if the dice show up {(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)}
- 8 if the dice show up {(2,6),(3,5),(4,4),(5,3),(6,2)}
- 9 if the dice show up {(3,6),(4,5),(5,4),(6,3)}
- 10 if the dice show up {(4,6),(5,5),(6,4)}
- 11 if the dice show up {(5,6),(6,5)}
- 12 if the dice show up {(6,6)}

x		2	3	4	5	6	7	8	9	10	11	12
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p(X = x)		$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$
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$$E(X) = \sum_{i=1}^{11} x_i p(x_i) = \frac{252}{36} = 7$$

Co-Variance:

If X and Y are two random variables, then the covariance between them is defined as:

$$\begin{aligned} \text{Cov}(X, Y) &= E[\{X - E(X)\}\{Y - E(Y)\}] \\ &= E[XY - X E(Y) - Y E(X) + E(X)E(Y)] \\ &= E(XY) - E(X) E(Y) - E(Y) E(X) + E(X)E(Y) \\ \text{Cov}(X, Y) &= E(XY) - E(X) E(Y) \end{aligned}$$

If X and Y are independent random variables, $E(XY) = E(X) E(Y)$
and hence, $\text{Cov}(X, Y) = E(X) E(Y) - E(X) E(Y) = 0$

Definition. The **correlation coefficient** between X and Y is defined to be

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{1}{\sigma_X \sigma_Y} \text{Cov}(X, Y)$$

In this definition we assume that $\text{Var}(X)$ and $\text{Var}(Y)$ are both finite and positive and that the square root in the denominator is the positive square root.

Properties:

$$\begin{aligned} 1. \quad \text{Cov}(aX, bY) &= E[\{aX - E(aX)\}\{bY - E(bY)\}] \\ &= E[a\{X - E(X)\} b\{Y - E(Y)\}] \\ &= ab E[\{X - E(X)\} \{Y - E(Y)\}] \\ &= ab \text{Cov}(X, Y) \end{aligned}$$

$$2. \quad \text{Cov}(X + a, Y + b) = \text{Cov}(X, Y)$$

$$3. \quad \text{Cov}\left(\frac{X - \bar{X}}{\sigma_X}, \frac{Y - \bar{Y}}{\sigma_Y}\right) = \frac{1}{\sigma_X \sigma_Y} \text{Cov}(X, Y)$$

4. If X and Y are independent, $\text{Cov}(X, Y) = 0$, but the converse is not true.

$$5. \quad V(X_1 \pm X_2) = V(X_1) + V(X_2) \pm 2\text{Cov}(X_1, X_2)$$

$$6. \quad V(\sum_{i=1}^n a_i X_i) = \sum_{i=1}^n a_i^2 V(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n a_i a_j \text{Cov}(X_i, X_j)$$

Example.

Two R.V's X and Y have joint pdf $f(x, y) = \begin{cases} \frac{xy}{96} & , 0 < x < 4, 1 < y < 5 \\ 0 & , elsewhere \end{cases}$

Find (i) E(X) (ii) E(Y) (iii) E(XY) (iv) E(2X + 3Y) (v) Var(X)

(vi) Cov(X, Y).

Solution:

$$\text{i) } E(X) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xf(x, y) dx dy$$

$$= \int_1^5 \int_0^4 x \left(\frac{xy}{96} \right) dx dy$$

$$= \frac{8}{3}$$

$$\text{ii) } E(Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} yf(x, y) dx dy$$

$$= \int_1^5 \int_0^4 y \left(\frac{xy}{96} \right) dx dy$$

$$= \frac{31}{9}$$

$$\text{iii) } E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x, y)dx dy$$

$$= \int_1^5 \int_0^4 xy \left(\frac{xy}{96} \right) dx dy$$

$$= \frac{248}{27}$$

$$\text{iv) } E[2X + 3Y] = 2E(X) + 3E(Y) = 2 \cdot \frac{8}{3} + 3 \cdot \frac{31}{9} = \frac{47}{3}$$

v) We know that, $Var(X) = E(X^2) - [E(X)]^2$

$$\text{Now, } E(X^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 f(x, y) dx dy$$

$$= \int_1^5 \int_0^4 x^2 \left(\frac{xy}{96} \right) dx dy$$

$$= 8$$

$$\Rightarrow Var(X) = E(X^2) - [E(X)]^2$$

$$= 8 - \left(\frac{8}{3} \right)^2 = \frac{8}{9}$$

$$\text{vi) } \text{Cov}(X, Y) = E(XY) - E(X).E(Y)$$

$$= \frac{248}{27} - \left(\frac{8}{3}\right)\left(\frac{31}{9}\right)$$

$$= 0$$

Example 6

Two random variables X and Y have the following joint probability density function:

$$f(x, y) = \begin{cases} 2 - x - y ; & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 ; & \text{otherwise} \end{cases}$$

- Find
- (i) Marginal probability density function of X and Y.
 - (ii) Conditional density function of X and Y
 - (iii) $\text{Var}(X)$, $\text{Var}(Y)$

Moment generating function

The moment generating function of a random variable X , denoted by $M_X(t)$ is defined by

► $M_X(t) = E(e^{tX}) = \sum_x e^{tx} P(X = x)$ when X is a discrete random variable,

► $M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$ when X is a continuous random variable.

Let X be a random variable with moment-generating function $M_X(t)$. Then

$$\left. \frac{d^r M_X(t)}{dt^r} \right|_{t=0} = \mu'_r.$$

Properties of MGF: If a, b are constants and X is a random variable, then

- $M_{aX}(t) = M_X(at),$
- $M_{X+a}(t) = e^{at} M_X(t),$
- $M_{\frac{X+a}{b}}(t) = e^{\frac{at}{b}} M_X(\frac{t}{b}),$
- $M_{\frac{X-a}{b}}(t) = e^{-\frac{at}{b}} M_X(\frac{t}{b}).$

If $M_X(t)$ and $M_Y(t)$ are the moment generating functions of the independent random variables X and Y , respectively, then

$$M_{X+Y}(t) = M_X(t)M_Y(t).$$

Exercise Suppose that a mathematician determines that the revenue the UConn Dairy Bar makes in a week is a random variable, X , with moment generating function

$$m_X(t) = \frac{1}{(1 - 2500t)^4}$$

Find the standard deviation of the revenue the UConn Dairy bar makes in a week.

Answer:

We have $SD(X) = \sqrt{\text{Var}(X)}$ and $\text{Var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2$.

Therefore we can compute

$$m'(t) = 4(2500)(1 - 2500t)^{-5},$$

$$m''(t) = 20(2500)^2(1 - 2500t)^{-6},$$

$$\mathbb{E}X = m'(0) = 10,000$$

$$\mathbb{E}X^2 = m''(0) = 125,000,000$$

$$\text{Var}(X) = 125,000,000 - 10,000^2 = 25,000,000$$

$$SD(X) = \sqrt{25,000,000} = \mathbf{5,000}.$$